Dairu Liu

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EDUCATION

Nankai University — Tianjin, China
B.Eng. in Software Engineering — *Sep. 2023 – Jun. 2027 (expected)*

The Affiliated High School of Peking University — Beijing, China

RESEARCH INTERESTS

My primary research goal is to leverage large-scale data to build **generalizable** embodied agents capable of performing diverse real world tasks. I am particularly interested in **Robot Learning** (scaling up robot data, efficient robot learning algorithms, humanoid whole-body control, loco-manipulation) and **Multimodal Reasoning** (agentic perception, cross-modal understanding, long-horizon decision making).

PUBLICATIONS & PREPRINTS

[1] **HumanTracker: Towards Comprehensive and Human-Aligned Motion Tracking Benchmark**
Dairu Liu*, Zekun Qi*, Jiayu Zeng*, Ruixi Yu, Yu Guan, Yintianrun Zhang, Xuchuan Chen, Sikai Liang, Zekai Li, Chenghuai Lin, Xinqiang Yu, Wenyao Zhang, He Wang†, Li Yi†
ECCV 2026 [Paper] [Project] [Code]

- Built a standardized humanoid motion tracking benchmark with ∼**153 hours** of newly captured optical mocap from **24 professional performers**, comprising **25K clips** organized into **4 motion families** for fine-grained diagnosis.
- Proposed **HumanScore**, a preference-aligned reward model trained on **12K pairwise human comparisons**; achieves **0.9083** alignment rate with human judgments, outperforming MPJPE (**0.8049**).

[2] **Humanoid-GPT: Scaling Data and Structure for Zero-Shot Motion Tracking**
Zekun Qi*, Xuchuan Chen*, **Dairu Liu***, Chenghuai Lin*, Yunrui Lian, Sikai Liang, Zhikai Zhang, Yu Guan, Jilong Wang, Wenyao Zhang, Xinqiang Yu, He Wang†, Li Yi†
CVPR 2026 [Paper] [Project] [Code]

- Introduced **Humanoid-GPT**, a GPT-style causal Transformer that scales humanoid motion tracking through large-scale data and structured policy learning, enabling robust zero-shot motion tracking on **Unitree G1**.
- Distilled hundreds of RL motion experts into one generalist tracker using DAgger and proposed **Harmonic Motion Embedding** (**HME**) for diversity-aware, distribution-balanced sampling over a ∼**2B-frame** retargeted motion corpus.

[3] **Evaluating Time Awareness and Cross-modal Active Perception of Large Models via 4D Escape Room Task**
Yurui Dong*, Ziyue Wang*, Shuyun Lu*, **Dairu Liu***, Xuechen Liu*, Fuwen Luo, Peng Li†, Yang Liu†
EMNLP 2026 [Paper] [Project]

- Developed **EscapeCraft-4D**, a customizable 4D escape-room benchmark (**66 scenes**, **1,032 objects**) incorporating trigger-based audio, misleading auditory distractors, and temporally transient clues for evaluating time-aware multimodal reasoning.
- Introduced **AMR** (Audio Misleading Rate) and **TCSS** (Transient Clue Seeing Speed) to measure auditory distraction and discovery of disappearing clues, revealing modality bias and failures in time-aware decision making.

[4] **Thinking with Visual Abstract: Enhancing Multimodal Reasoning via Visual Abstraction**
Dairu Liu*, Ziyue Wang*, Minyuan Ruan*, Fuwen Luo, Chi Chen, Peng Li†, Yang Liu†
NeurIPS 2025 Workshop [Paper] [Project]

- Proposed **Visual Abstract Thinking** (**VAT**), a reasoning paradigm that replaces verbose verbal CoT with visual abstractions, focusing models on semantic, structural, and geometric cues.
- VAT achieves +**2.21%** average gain over the GPT-5 baseline, exceeding CoT (+**0.96%**); uses only **0.68**× CoT runtime and **0.1**× the tokens of Visual Sketchpad on GPT-4o. RL-trained VAT further gains +**4.52%** on Qwen-2.5-VL-3B.

[5] **LIMMT: Less is More for Motion Tracking**
Yu Guan*, Zekun Qi*, Chenghuai Lin, Xuchuan Chen, **Dairu Liu**, Wenyao Zhang, Jilong Wang, Xinqiang Yu, He Wang†, Li Yi†
ICML 2026 [Paper]

- Conducted the first data-centric study for humanoid motion tracking, showing that carefully selected compact motion subsets can outperform training on the full motion corpus.
- Proposed **General Quality Selection** (**GQS**) using physics feasibility, diversity, and complexity; training on only ∼**3%** of AMASS outperforms the full corpus and transfers to real-world **Unitree G1** deployment.

[6] **Where is Physical RSI capability from? An Observational Scaling Law of LLM-as-policy**
Zekai Li*, Jiayi Yu*, Jinlu Zhang*, Luyang Niu*, Zekun Qi*, Jun Zheng, Xuchuan Chen, **Dairu Liu**, Yintianrun Zhang, He Wang†, Li Yi†
arXiv Preprint, 2026 [Paper] [Project]

- Evaluated physical recursive self-improvement on **18 manipulation tasks** with **PhysRSI-eval**; studied capability scaling, prediction from early feedback, and transfer of retained experience.

[7] **Systematically Exploring the Capabilities of GPT-6 Astra as Embodied Policies**
Galbot Team (including **Dairu Liu**)
Tech Report, 2026 [Paper]

- **Core contributor to the humanoid section**, studying GPT-6 Astra as an embodied policy for humanoid loco-manipulation with pretrained whole-body controllers.

RESEARCH EXPERIENCE

Research Intern, GALBOT Beijing, China
Advisor: **Prof. Li Yi**, **Prof. He Wang** *Dec. 2025 – Present*

- **S2–S1: Humanoid VLA.** Developed vision-language-action policies for humanoid **loco-manipulation** and explored their integration with high-level reasoning.
- **Egocentric data.** Explored the use of human ego data in humanoid robot learning and VLA policy training.
- **S0: AstraBrain-WBC.** Explored training a humanoid whole-body control foundation model as a low-level cerebellum for higher-level policies.

Research Intern, THUNLP | AIR, Tsinghua University Beijing, China
Advisor: **Prof. Peng Li**, **Prof. Yang Liu** *Apr. 2024 – Jan. 2026*

- Proposed Visual Abstract Thinking, a novel multimodal reasoning paradigm that enhances MLLMs via visual abstraction rather than verbose verbal rationales. [4]
- Developed a 4D escape-room benchmark for evaluating time awareness and cross-modal active perception in Omni models. [3]

HONOURS & AWARDS

National Scholarship (Top 0.2%; Undergraduate, 10000CNY) 2026
National Scholarship (Top 0.2%; Undergraduate, 10000CNY) 2024
Second Prize, NOIP 2018 (National Olympiad in Informatics in Provinces) Dec. 2018
Second Prize, CSP-S 2019 (CCF Certified Software Professional) Dec. 2019

LEADERSHIP & SERVICES

Vice President, Quantitative Finance & Investment Society, Nankai University Sep. 2025 – Present
Member, University Ski Team, Nankai University 2024 – 2025
Academic Department Vice-Head, Psychology Association, Nankai University Sep. 2024 – Sep. 2025

SKILLS

Robot Platforms: Unitree G1
Programming: C++, Python, MATLAB, LaTeX
Languages: Mandarin Chinese (native); English (IELTS 7.0 – L 7.5, R 8.0, S 6.0, W 6.0; CET-6 608)
Interests: Skiing